Mayank Kumar

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Professional Summary

Machine Learning Engineer with 3+ years of experience building real-time, low-latency ML systems at scale, including anomaly detection, behavioral analytics, and classification pipelines in production. Independently designed and built an end-to-end agentic AI system integrating retrieval-augmented generation, multi-provider LLM orchestration, and multimodal analysis. Published researcher in adversarial machine learning and federated learning security.

Technical Skills

Languages: Python, R, C++, C, SQL
Classical ML: LightGBM, Isolation Forest, Random Forest, Naive Bayes, SVM, Scikit-learn
Deep Learning & NLP: PyTorch, TensorFlow, Keras, LSTM, CNN, NLTK, GloVe, NLP, OpenCV
Agentic AI / LLM: Claude API, Ollama, Model Context Protocol (MCP), RAG, LLM Evaluation, Prompt Engineering
Web Frameworks: Django, Flask, FastAPI, React
Infra & Data: Docker, Apache Kafka, PostgreSQL, OpenSearch, ChromaDB, SQLite, REST APIs, Git, Linux

Work Experience

Centre for Development of Telematics (C-DOT) — Delhi, India
Scientist - B (Machine Learning Engineer) — July 2023 – Present

- Designed and deployed an endpoint-compatible malware scoring pipeline using static PE/ELF analysis and LightGBM, optimized for low-latency pre-execution detection and risk-based decision making.
- Built User and Entity Behavior Analytics (UEBA) features to detect access and authentication anomalies, including off-hour and abnormal activity, using unsupervised learning.
- Reduced false positives by 90% by replacing a rule-based anomaly detection system with an unsupervised learning approach, and demonstrated superior performance of Isolation Forest compared to One-Class SVM on internal datasets.
- Extended UEBA capabilities by integrating system workload behavior analysis, enabling detection of abnormal CPU, memory, and disk usage patterns across monitored endpoints.
- Developed real-time ML inference pipelines using FastAPI, Docker, and Kafka to collect and process telemetry data from 10,000+ endpoints, enabling low-latency anomaly detection and continuous model training at scale.
- Owned the full ML lifecycle from feature engineering to production deployment, implementing versioning and monitoring workflows to track model performance and drift.
- **Technology Stack**: Python, LightGBM, Isolation Forest, FastAPI, Docker, Apache Kafka, PostgreSQL, OpenSearch, REST APIs, Linux, Git

Indian Institute of Technology (IIT), Mandi — Mandi, India
AI Research Intern — May 2022 - Jul 2022

- **Spoken Language Identification (Sponsored by National Language Translation Mission)**:
  * Developed an X-vector system using MFCC for feature extraction with a training accuracy of 96%, and
  * Created a GUI for live demonstration using PyTorch, Pygame, CUDA, Numpy, and Keras.

Projects

Stock Agent | Agentic AI System for Real-Time Stock Market Analysis & Trading Research [GitHub]: Ongoing

- Built an end-to-end agentic AI system integrating live NSE market data, news retrieval, and personal-document search (RAG) into a conversational agent, accessible via MCP (Claude Desktop) and a custom full-stack web application, with a shared tool layer across both interfaces.
- Implemented a dual-provider LLM architecture (Claude API + local Ollama) with live per-message provider switching; designed and ran a systematic benchmark comparing local vs. cloud models across tool-routing accuracy, output-format compliance, and multi-field numerical reasoning, surfacing a non-obvious capability gap between correct tool usage and correct data interpretation.
- Engineered a RAG pipeline with vector search over a 29,000+ chunk personal knowledge base (20+ PDF/EPUB documents), generating local embeddings and querying a vector database, with OCR fallback for scanned/corrupted files, achieving fully offline operation at zero per-query cost.
- Built multimodal chart-image analysis with automatic routing between cloud and local vision models; deployed the application (FastAPI + React, streaming SSE) to production and diagnosed a real third-party data-provider infrastructure limitation.
- **Technology Stack:** Python, FastAPI, React, Anthropic Claude API, Ollama, MCP, RAG, ChromaDB, sentence-transformers, SQLite, Tesseract OCR, Render, Vercel

Researched the resistance of Split Neural Networks and Federated Learning against Adversarial Attacks.:

- Conducted an experiment using the CIFAR-10 dataset with the ResNet18 model, designing a data poisoning attack and white-box attack against Split Learning and Federated Learning systems.
- Demonstrated decreased model accuracy and target-class misclassification by poisoning training data and adding adversarial perturbations to test images.
- **Technology Stack:** PyTorch, Pygame, GANs,

Publications

Conference: Sharma, M., Kumar, M., Gonzalez, C., Dutt, V. How does the Presence of Cognitive Biases in Phishing Emails Affect Human Decision-Making? (Paper published in ICONIP 2022) [Paper]

Conference: Kumar, M., Aggrawal, R., Singh, P. BATFL: Battling Backdoor Attacks in Federated Learning (Paper published in Conference SINCOF 2023) [Paper]

Conference: Kumar, M., Aggrawal, R., Singh, P. RAFT: Evaluating Federated Learning Resilience Against Threats (Paper published in Conference SINCOF 2023) [Paper]

Conference: Kumar, M., Singh, P., Gupta, S. On Robustness of Split Neural Networks Against Data Poisoning Attacks (Paper accepted in Conference IJCNN 2023) [Paper]

Education

Indian Institute of Technology (IIT), Jammu — Jammu, India
M.Tech, CSE (CGPA: 8.77) — July 2021 - June 2023

Government Engineering College, Jhalawar — Rajasthan, India
B.Tech (Hons.), CSE (Percentage: 71.76%) — July 2016 - June 2020

Professional Development

- Agentic AI with LangChain and LangGraph (Issued by: IBM - Coursera)
- Build RAG Applications: Get Started (Issued by: IBM: Coursera)
- Develop Generative AI Applications: Get Started (Issued by:IBM: Coursera)
- Linear Algebra for Machine Learning and Data Science (Issued by: DeepLearning.AI, Coursera)
- Neural Network And Deep Learning (Issued by: DeepLearning.AI: Coursera)